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1 sum kth powers

The first seven examples of Faulhaber's formula are

$$\sum_{k=1}^{n} k^0 = \frac{1}{1}\left(n\right)$$
$$\sum_{k=1}^{n} k^1 = \frac{1}{2}\left(n^2 + \tfrac{2}{2}n\right)$$
$$\sum_{k=1}^{n} k^2 = \frac{1}{3}\left(n^3 + \tfrac{3}{2}n^2 + \tfrac{3}{6}n\right)$$
$$\sum_{k=1}^{n} k^3 = \frac{1}{4}\left(n^4 + \tfrac{4}{2}n^3 + \tfrac{6}{6}n^2 + 0n\right)$$
$$\sum_{k=1}^{n} k^4 = \frac{1}{5}\left(n^5 + \tfrac{5}{2}n^4 + \tfrac{10}{6}n^3 + 0n^2 - \tfrac{5}{30}n\right)$$
$$\sum_{k=1}^{n} k^5 = \frac{1}{6}\left(n^6 + \tfrac{6}{2}n^5 + \tfrac{15}{6}n^4 + 0n^3 - \tfrac{15}{30}n^2 + 0n\right)$$
$$\sum_{k=1}^{n} k^6 = \frac{1}{7}\left(n^7 + \tfrac{7}{2}n^6 + \tfrac{21}{6}n^5 + 0n^4 - \tfrac{35}{30}n^3 + 0n^2 + \tfrac{7}{42}n\right).$$

The coefficients of Faulhaber's formula in its general form involve the Bernoulli numbers B_j. The Bernoulli numbers begin

$$B_0 = 1 \quad B_1 = \tfrac{1}{2} \quad B_2 = \tfrac{1}{6} \quad B_3 = 0$$
$$B_4 = -\tfrac{1}{30} \quad B_5 = 0 \quad B_6 = \tfrac{1}{42} \quad B_7 = 0,$$

where here we use the convention that $B_1 = +\frac{1}{2}$. The Bernoulli numbers have various definitions (see Bernoulli number § Definitions), such as that they are the coefficients of the exponential generating function

$$\frac{t}{1-\mathrm{e}^{-t}} = \frac{t}{2}\left(\coth\frac{t}{2} + 1\right) = \sum_{k=0}^{\infty} B_k \frac{t^k}{k!}.$$

Then Faulhaber's formula is that

$$\sum_{k=1}^{n} k^p = \frac{1}{p+1}\sum_{r=0}^{p} \binom{p+1}{r} B_r n^{p+1-r}.$$

Извлечение квадратного корня по простому модулю

Алгоритм:
`input:` $3 \leqslant p$ — простое, $1 \leqslant a < p$
`output:` $1 \leqslant x < p$ такой, что $x^2 = a$ или -1, если $\nexists x$

1. Если $a^{\frac{p-1}{2}} \neq 1$ вернуть $x = -1$
2. Пока корень не найден:
3. $\quad i$:= `random` $[1..p-1]$
4. $\quad T(x) := (x+i)^{\frac{p-1}{2}} - 1 \mod (x^2 - a) = b \cdot x + c$ (многочлены от x)
5. $\quad$ Если $b \neq 0$, то вернуть $x = c \cdot b^{-1} = c \cdot b^{p-2}$

2 ******** general matching

```
/* _mm_setcsr( _mm_getcsr() | 0x0040 | 0x8000 ); */ #include <bits/stdc++.h>
std::vector<std::pair<int, int>> maximum_matching(int n, const std::vector<std::pair<int, int>>
    &edg) {
    struct DSU { std::vector<int> prv;
        DSU(int n = 0) : prv(n, -1) { ; }
        void clear(int n) { prv.assign(n, -1); }
        int get(int i) { return prv[i] == -1 ? i : prv[i] = get(prv[i]); }
    }; std::vector<int> vec(n, -1); std::vector<std::vector<int>> gr(n); std::vector<int>
        cnt(n); for (auto [u, v] : edg) cnt[u]++, cnt[v]++; for (int i = 0; i < n; i++)
        gr[i].reserve(cnt[i]); for (auto [u, v] : edg) gr[u].push_back(v), gr[v].push_back(u);
        DSU dsu(n), dsu2(n + 1); std::vector<std::array<int, 2>> prv(n, {-1, -1});
        std::vector<std::array<int, 3>> fwd(n, {-1, -1, -1}), bwd(n, {-1, -1, -1});
        std::vector<int> bl_dir(n, -1), bl_prv(n, -1), bl_lvl(n, -1); std::vector<int>
        bl_depth(n, -1), bl_jump(n, -1), bl_ord(n, -1); std::vector<int> depth(n, -1);
        std::vector<int64_t> used(n, -1); int64_t used_mark = -1; std::deque<int> deque;
        std::vector<std::pair<int, int>> aug_vec; int total_bl = 0; auto find_lca = [&](int u,
        int v) { used_mark++, u = dsu.get(u), v = dsu.get(v); while (u >= 0 || v >= 0) { if (u
        != -1) { if (used[u] == used_mark) return u; used[u] = used_mark, (u = prv[u][0] >= 0 ?
        dsu.get(prv[u][0]) : prv[u][0]); } std::swap(u, v); }
        return -1; }; auto contract_blossom = [&](int u0, int v0, int l) { bl_lvl[l] =
            ++total_bl, bl_ord.push_back(l); for (int ch = 0; ch < 2; ch++) { int x0 = ch == 0 ?
            u0 : v0, y0 = ch == 0 ? v0 : u0; int x = dsu.get(x0); while (x != l) { int p0 =
            prv[x][0]; int p = dsu.get(p0); if (depth[x] == 1) deque.push_back(x); bl_dir[x] =
            depth[x] ^ ch, bl_prv[x] = l; fwd[x] = {dsu.get(p0), p0, prv[x][1]}, bwd[x] =
            {dsu.get(y0), y0, x0}; if (ch) std::swap(fwd[x], bwd[x]); x0 = p0, y0 = prv[x][1], x
            = p; } }
        for (auto x : std::array<int, 2>{dsu.get(u0), dsu.get(v0)}) { for (; x != l; x =
            dsu.get(prv[x][0])) { dsu.prv[x] = l; }}
    }; auto init_jump = [&] { for (int i = bl_ord.size() - 1; i >= 0; i--) if (int v =
        bl_ord[i]; bl_depth[v] == -1) { int f = bl_prv[v]; bl_depth[v] = bl_depth[f] + 1,
        bl_jump[v] = f; if (bl_depth[v] > 1 && bl_depth[f] - bl_depth[bl_jump[f]] ==
        bl_depth[bl_jump[f]] - bl_depth[bl_jump[bl_jump[f]]]) { bl_jump[v] =
        bl_jump[bl_jump[f]]; } }
    }; auto jump = [&](int x, int d) { while (bl_lvl[bl_prv[x]] < d) { if (bl_lvl[bl_jump[x]] <
        d) { x = bl_jump[x]; } else { x = bl_prv[x]; } }
        return x; }; auto augment = [&](int u0, int v0) { int u = dsu.get(u0), v = dsu.get(v0);
            std::list<std::pair<int, int>> list; for (int x = dsu.get(u0); prv[x][0] != -1; x =
            dsu.get(prv[x][0])) { list.push_front({prv[x][0], prv[x][1]}); }
        list.push_front({-1, prv[u][0] == -1 ? u : dsu.get(list.front().first)});
            list.push_back({u0, v0}); for (int x = dsu.get(v0); prv[x][0] != -1; x =
            dsu.get(prv[x][0])) { list.push_back({prv[x][1], prv[x][0]}); }
        list.push_back({prv[v][0] == -1 ? v : dsu.get(list.back().second), -1}); for (auto it =
            std::next(list.begin()); it != list.end();) { auto [a1, a2] = *std::prev(it); auto
            [b1, b2] = *it; if (a2 == b1) { it++; continue; } bool f1 = (a1 == -1 || vec[a1] ==
            a2), f2 = (b2 == -1 || vec[b1] == b2); assert(f1 != f2); bool rev = false; if (f2)
            rev = true, std::swap(a1, b2), std::swap(a2, b1); decltype(list) list2; int c =
            jump(b1, bl_lvl[a2]); auto &mv = !bl_dir[c] ? fwd : bwd; for (int x = c; x != a2; x
            = mv[x][0]) list2.push_front({mv[x][1], mv[x][2]}); if (!rev) { it = list.insert(it,
            list2.begin(), list2.end()); } else { for (auto &[a, b] : list2) std::swap(a, b); it
            = list.insert(it, list2.rbegin(), list2.rend()); } }
        list.pop_front(), list.pop_back(); assert(list.size() % 2 == 1); for (auto it =
            list.begin();; it = std::next(it, 2)) { vec[it->first] = it->second, vec[it->second]
            = it->first; if (std::next(it) == list.end()) break; }
    }; auto process_edge = [&](int u0, int v0) { int u = dsu.get(u0), v = dsu.get(v0); if (u ==
        v || depth[v] == 1 || dsu2.get(u) == n || dsu2.get(v) == n) return; if (depth[v] == -1)
        { int w = vec[v]; dsu2.prv[v] = u, dsu2.prv[w] = u, depth[v] = 1; depth[w] = 0, prv[v] =
        {u0, v0}, prv[w] = {v0, w}; deque.push_back(w); } else { int l = find_lca(u, v); if (l
        != -1) { contract_blossom(u0, v0, l); } else { /* augment(u0, v0); */
        aug_vec.push_back({u0, v0}); for (int x : std::array<int, 2>{u, v}) { for (; x != -1; x
            = (prv[x][0] == -1 ? -1 : dsu.get(prv[x][0]))) { dsu2.prv[x] = n; } } } }
    }; while (true) { dsu.clear(n), dsu2.clear(n + 1), prv.assign(n, {-1, -1}); fwd.assign(n,
        {-1, -1, -1}), bwd.assign(n, {-1, -1, -1}); bl_dir.assign(n, -1), bl_prv.assign(n, -1),
        bl_lvl.assign(n, -1); depth.assign(n, -1), deque.clear(), aug_vec.clear();
        bl_ord.assign(n, -1), bl_depth.assign(n, -1), bl_jump.assign(n, -1);
        std::iota(bl_ord.begin(), bl_ord.end(), 0), std::iota(bl_prv.begin(), bl_prv.end(), 0);
        total_bl = 0; for (int i = 0; i < n; i++) { if (vec[i] == -1) depth[i] = 0, bl_lvl[i] =
        total_bl, deque.push_back(i); }
        while (deque.size()) { int u0 = deque.front(); deque.pop_front(); for (int v0 : gr[u0])
            process_edge(u0, v0); }
        if (aug_vec.size()) { init_jump(); std::cerr << aug_vec.size() << " " ; for (auto [u0,
            v0] : aug_vec) augment(u0, v0); } else { break; }
    }
    std::vector<std::pair<int, int>> res; for (int i = 0; i < n; i++) if (vec[i] > i)
        res.push_back({i, vec[i]}); return res; }
```

3 svg

```
#include <stdint.h> #include <fstream> #include <sstream>
template <typename T>
struct Vector { T x, y;
    Vector() : x(0), y(0) { ; }
    Vector(T x, T y) : x(x), y(y) { ; }
    /* template <typename T2> */
    /* operator Vector<T2>() const { */ /* return Vector<T2>(x, y); */ /* } */
    Vector operator+(const Vector& other) const { return Vector(x + other.x, y + other.y); }
    Vector operator-() const { return Vector(-x, -y); }
    Vector operator-(const Vector& other) const { return Vector(x - other.x, y - other.y); }
    T operator*(const Vector& other) const { return x * other.x + y * other.y; }
    T operator%(const Vector& other) const { return x * other.y - y * other.x; }
    Vector<double> operator*(const double& val) const { return Vector(x * val, y * val); }
    bool operator==(const Vector& other) const { return x == other.x && y == other.y; }
};
struct Svg { std::stringstream sout; static constexpr double scale = 500; static constexpr
    double shift = 50;
    Svg() { clear(); }
    void clear() { sout = std::stringstream(); sout.precision(5); sout << std::fixed; sout <<
        R"meow(<svg width=\"1000px\" height=\"1000px\" style=\"background-color:lightgreen\"
        xmlns=\ "http://www.w3.org/2000/svg\">\n)meow" ; }
    void print() { std::string s = sout.str(); s += "</svg>\n" ; std::ofstream fout( "meow.svg"
        ); fout << s << "\n" ; fout.flush(); fout.close(); }
    void line(Vector<double> pt1, Vector<double> pt2, std::string color, double width = 1) {
        sout << "<line " ; sout << "x1=\"" << (float)(pt1.x * scale + shift) << "\" " ; sout <<
        "y1=\"" << (float)((scale - pt1.y * scale) + shift) << "\" " ; sout << "x2=\"" <<
        (float)(pt2.x * scale + shift) << "\" " ; sout << "y2=\"" << (float)((scale - pt2.y *
        scale) + shift) << "\" " ; sout << " stroke=\"" << color << "\"" ; sout <<
        " stroke-width=\"" << float(width) << "\"" ; sout << "/>\n" ; }
    void circle(Vector<double> pt, double r, std::string color, double width = 1) { sout <<
        "<circle " ; sout << "cx=\"" << float(pt.x * scale + shift) << "\" " ; sout << "cy=\""
        << float((scale - pt.y * scale) + shift) << "\" " ; sout << "r=\"" << float(r) << "\" "
        ; sout << " stroke=\"" << color << "\"" ; sout << " stroke-width=\"" << float(width) <<
        "\"" ; sout << "/>\n" ; }; };
```

4 floor sum + mod linear

```
/* forged from mangooste.ru/lib/algo/mod-of-linear */ /* ! WARNING: careful with overflow. Don't
    forget to specify large enough type T. * Returns sum_{x = 0}^{n - 1} floor((kx + b) / m). *
    Require: k >= 0, b >= 0, m > 0, n >= 0. */
template <typename T>
```

```
T floor_sum(T k, T b, T m, T n) { if (k == 0) { return (b / m) * n; } if (k >= m || b >= m) {
    return n * (n - 1) / 2 * (k / m) + n * (b / m) + floor_sum(k % m, b % m, m, n); } T ymax =
    (k * (n - 1) + b) / m; return n * ymax - floor_sum(m, m + k - b - 1, k, ymax); }
/* ! WARNING: careful with overflow. Don't forget to specify large enough to fit floor_sum type
    T. * Returns sum_{x = 0}^{n - 1} (kx + b) % m. * Require: m > 0, n >= 0. */
template <typename T>
T mod_sum(T k, T b, T m, T n) { k = (k % m + m) % m; b = (b % m + m) % m; return n * (n - 1) / 2
    * k + n * b - m * floor_sum(k, b, m, n); }
/* --------- */ /* * Returns min_{x=0}^{n - 1} (kx + b) mod m * Require: n, m > 0, 0 <= b, k < m
    */
template <typename T> T min_of_mod_of_linear(T n, T m, T k, T b, T step_cost = 1, T
    overflow_cost = 0);
/* * Returns max_{x=0}^{n - 1} (kx + b) mod m * Require: n, m > 0, 0 <= b, k < m */
template <typename T> T max_of_mod_of_linear(T n, T m, T k, T b, T step_cost = 1, T
    overflow_cost = 0);
template <typename T>
T max_of_mod_of_linear(T n, T m, T k, T b, T step_cost, T overflow_cost) { if (k == 0) { return
    b; } if (b < m - k) { T steps = (m - b - 1) / k; T cost = step_cost * steps; if (cost >= n)
    { return k * ((n - 1) / step_cost) + b; } n -= cost; b += steps * k; }
    return m - 1 - min_of_mod_of_linear( n, k, m % k, m - 1 - b, (m / k) * step_cost +
        overflow_cost, step_cost); }
template <typename T>
T min_of_mod_of_linear(T n, T m, T k, T b, T step_cost, T overflow_cost) { if (k == 0) { return
    b; } if (b >= k) { T steps = (m - b + k - 1) / k; T cost = step_cost * steps +
    overflow_cost; if (cost >= n) { return b; } n -= cost; b += steps * k - m; }
    return k - 1 - max_of_mod_of_linear( n, k, m % k, k - 1 - b, (m / k) * step_cost +
        overflow_cost, step_cost); }
```

5 primality test + Rho factorization

```
#include <bits/stdc++.h>
using u64 = uint64_t;
using u128 = __uint128_t;
u64 add(u64 a, u64 b, u64 mod) { return a + b - mod * (a + b >= mod); }
u64 mul(u64 a, u64 b, u64 mod) { return u128(a) * b % mod; }
u64 power(u64 b, u64 e, u64 mod) { u64 r = 1; for (; e > 0; e >>= 1) { if (e & 1) { r = mul(r,
    b, mod); } b = mul(b, b, mod); } return r; }
u64 is_prime(u64 val) { if (val % 2 == 0) { return val == 2; } int lg = __builtin_ctzll(val -
    1); for (u64 a : std::array{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 39, 41}) { u64 pw =
    power(a, val - 1 >> lg, val); for (int i = 0; i < lg && pw != 1; i++) { u64 pw2 = mul(pw,
    pw, val); if (i + 1 == lg && pw2 != 1) { return false; } if (pw2 == 1 && pw != val - 1) {
    return false; } pw = pw2; } } return true; }
u64 find_divisor(u64 val) { for (int i = 2; i <= 1000; i++) { if (val % i == 0) { if (val == i)
    { return 0; } return i; } } if (val <= 1e6 || is_prime(val)) { return 0; } auto f = [&](u64
    x) { return add(mul(x, x, val), 3, val); }; static std::mt19937_64 rnd; u64 a = rnd() % val;
    u64 b = a; std::vector<int64_t> vec; for (int64_t it = 0;; it++) { a = f(a); b = f(f(b));
    u64 diff = std::max(a, b) - std::min(a, b); vec.push_back(diff); if (vec.size() >= 200) {
    u64 prod = 1; for (u64 i : vec) { if (prod != 0) { prod = mul(prod, i, val); } } if
    (std::gcd(prod, val) != 1) { for (u64 i : vec) { if (std::gcd(i, val) != 1) { return
    std::gcd(i, val); } } assert(false); } vec.clear(); } } }
```

6 tree bitset

```
#include <iostream>
#pragma GCC optimize( "O3" )
#pragma GCC target( "avx2,lzcnt,bmi,bmi2" ) #include <bits/stdc++.h>
template <typename u_tp = uint64_t>
class TreeBitset { private: static constexpr size_t B = sizeof(u_tp) * 8; std::vector<u_tp>
    my_data; std::vector<u_tp*> data; size_t n, lg; public:
    TreeBitset(size_t n = 0) { assign(n); }
    void assign(size_t n) { this->n = n; size_t m = n + 2; std::vector<size_t> vec; while (m >
        1) { m = (m - 1) / B + 1; vec.push_back(m); } std::reverse(vec.begin(), vec.end()); lg =
        vec.size(); data.resize(vec.size()); size_t sum = std::accumulate(vec.begin(),
        vec.end(), size_t(0)); my_data.assign(sum, 0); for (size_t i = 0, s = 0; i < lg; s +=
        vec[i], i++) { data[i] = my_data.data() + s; } for (size_t i = 0, k = lg; k--; i /= B) {
        data[k][i / B] |= u_tp(1) << i % B; } for (size_t i = n + 1, k = lg; k--; i /= B) {
        data[k][i / B] |= u_tp(1) << i % B; } }
    size_t size() const { return n; }
    void clear() { my_data.assign(my_data.size(), 0); for (size_t i = 0, k = lg; k--; i /= B) {
        data[k][i / B] |= u_tp(1) << i % B; } for (size_t i = n + 1, k = lg; k--; i /= B) {
        data[k][i / B] |= u_tp(1) << i % B; } }
    /* i must be in [0, n) */
    bool insert(size_t i) { i++; if ((data[lg - 1][i / B] >> i % B) & 1) { return false; } for
        (size_t k = lg; k--; i /= B) { data[k][i / B] |= u_tp(1) << i % B; } return true; }
    /* i must be in [0, n) */
    bool erase(size_t i) { i++; if (!((data[lg - 1][i / B] >> i % B) & 1)) { return false; }
        data[lg - 1][i / B] ^= u_tp(1) << i % B; i /= B; for (size_t k = lg - 1; k > 0 &&
        !data[k][i]; k--, i /= B) { data[k - 1][i / B] ^= u_tp(1) << i % B; } return true; }
    /* i must be in [0, n) */
    bool contains(size_t i) const { i++; return (data[lg - 1][i / B] >> i % B) & 1; }
    /* i must be in [0, n] */ /* smallest element greater than or equal to i, n if doesn't exist
        */
    size_t find_next(size_t i) const { i++; size_t k = lg - 1; for (; !u_tp(data[k][i / B] >> i
        % B); k--) { i = i / B + 1; } for (; k < lg; k++) { u_tp mask = u_tp(data[k][i / B] >> i
        % B) << i % B; size_t ind = std::countr_zero(mask); i = (i / B * B + ind) * B; } i /= B;
        return i - 1; }
    /* i must be in [0, n) */ /* largest element less than or equal to i, n if doesn't exist */
    size_t find_prev(size_t i) const { i++; size_t k = lg - 1; for (; !u_tp(data[k][i / B] << (B
        - i % B - 1)); k--) { i = i / B - 1; } for (; k < lg; k++) { u_tp mask = u_tp(data[k][i
        / B] << (B - i % B - 1)) >> (B - i % B - 1); assert(mask); size_t ind = B - 1 -
        std::countl_zero(mask); i = (i / B * B + ind) * B + (B - 1); } i /= B; if (i == 0) {
        return n; } return i - 1; }
};
```

7 link cut

```
#include <bits/stdc++.h>
struct LinkCut {
    struct Node { int next[2], prev, sz, flip;
        Node() : next{-1, -1}, prev(-1), sz(1), flip(0) { ; }
    }; std::vector<Node> data;
    LinkCut(int n = 0) : data(n) { ; }
    int get_sz(int i) { return i == -1 ? 0 : data[i].sz; }
    void set_prev(int i, int p) { i != -1 ? data[i].prev = p : 0; }
    void flip(int i) { i != -1 ? data[i].flip ^= 1 : 0; }
    void pull(int i) { data[i].sz = get_sz(data[i].next[0]) + 1 + get_sz(data[i].next[1]); }
    void push(int i) {
        data[i].flip ? (flip(data[i].next[0]), flip(data[i].next[1]), std::swap(data[i].next[0],
            data[i].next[1]), data[i].flip = false) : 0; }
    int ch_num(int i) { return data[i].prev == -1 ? -1 : (data[data[i].prev].next[0] == i ? 0 :
        (data[data[i].prev].next[1] == i ? 1 : -1)); }
    void rotate(int i) { int j = data[i].prev, k = data[j].prev, ni = ch_num(i), nj = ch_num(j);
        data[j].next[ni] = data[i].next[!ni], set_prev(data[i].next[!ni], j); data[i].next[!ni]
        = j, data[i].prev = k, data[j].prev = i, pull(j), pull(i); if (nj != -1)
        data[k].next[nj] = i; }
    void splay(int i) { push(i); while (ch_num(i) != -1) { int j = data[i].prev, k =
        data[j].prev; if (ch_num(j) == -1) { push(j), push(i), rotate(i); } else { push(k),
        push(j), push(i), rotate(ch_num(i) == ch_num(j) ? j : i), rotate(i); } } }
    void expose(int i) { splay(i), data[i].next[1] = -1, pull(i); while (data[i].prev != -1) {
```

```
        splay(data[i].prev), data[data[i].prev].next[1] = i, pull(data[i].prev), splay(i); } }
    void ch_root(int i) { expose(i), flip(i); }
    void link(int i, int j) { ch_root(j), data[j].prev = i; }
    void cut(int i, int j) { ch_root(i), expose(i), splay(j), data[j].prev = -1; }
    int get_dist(int i, int j) { return i == j ? 0 : (ch_root(i), expose(j), (data[i].prev != -1
        ? data[j].sz - 1 : -1)); }
};
```

8 simplex

```
/* src: https://github.com/dacin21/dacin21_codebook/blob/master/numerical/simplex_lp.cpp */ /*
    Two-phase simplex algorithm for solving linear programs of the form */ /* */ /* maximize c^T
    x */ /* subject to Ax <= b */ /* x >= 0 */
/* OUTPUT: value of the optimal solution (inf if unbounded */ /* above, -inf if infeasible) */
    /* */ /* To use this code, create an LPSolver object with A, b, and c as */ /* arguments.
    Then, call Solve(x). */ /* 2018-05: added steepest edge pricing, speed up factor of ~5 */
template <typename DOUBLE>
struct Simplex_Steep {
    using VD = vector<DOUBLE>;
    using VVD = vector<VD>;
    using VI = vector<int>; const DOUBLE EPS = 1e-12; int m, n; VI B, N; VVD D; int
        iteration_cnt = 0;
    Simplex_Steep(const VVD& A, const VD& b, const VD& c) : m(b.size()), n(c.size()), B(m), N(n
        + 1), D(m + 2, VD(n + 2)) { for (int i = 0; i < m; i++) for (int j = 0; j < n; j++)
        D[i][j] = A[i][j]; for (int i = 0; i < m; i++) { B[i] = n + i; D[i][n] = -1; D[i][n + 1]
        = b[i]; } for (int j = 0; j < n; j++) { N[j] = j; D[m][j] = -c[j]; } N[n] = -1; D[m +
        1][n] = 1; }
    void Pivot(int r, int s) { for (int i = 0; i < m + 2; i++) if (i != r) for (int j = 0; j < n
        + 2; j++) if (j != s) D[i][j] -= D[r][j] * D[i][s] / D[r][s]; for (int j = 0; j < n + 2;
        j++) if (j != s) D[r][j] /= D[r][s]; for (int i = 0; i < m + 2; i++) if (i != r) D[i][s]
        /= -D[r][s]; D[r][s] = 1.0 / D[r][s]; swap(B[r], N[s]); }
    bool Simplex(int phase) { int x = phase == 1 ? m + 1 : m; while (true) { ++iteration_cnt;
        int s = -1; DOUBLE c_val = -1; for (int j = 0; j <= n; j++) { if (phase == 2 && N[j] ==
        -1) continue; DOUBLE norm_sq = 0; for (int k = 0; k <= m; ++k) { norm_sq += D[k][j] *
        D[k][j]; } if (norm_sq < EPS) norm_sq = EPS; /* stop division by 0 */ DOUBLE c_val_j =
        D[x][j] / sqrtl(norm_sq); if (s == -1 || c_val_j < c_val || (c_val == c_val_j && N[j] <
        N[s])) { s = j; c_val = c_val_j; } } if (D[x][s] >= -EPS) return true; int r = -1; for
        (int i = 0; i < m; i++) { if (D[i][s] <= EPS) continue; if (r == -1 || D[i][n + 1] /
        D[i][s] < D[r][n + 1] / D[r][s] || (D[i][n + 1] / D[i][s] == D[r][n + 1] / D[r][s] &&
        B[i] < B[r])) r = i; } if (r == -1) return false; Pivot(r, s); } }
    DOUBLE solve(VD& x) { int r = 0; for (int i = 1; i < m; i++) if (D[i][n + 1] < D[r][n + 1])
        r = i; if (D[r][n + 1] <= -EPS) { Pivot(r, n); if (!Simplex(1) || D[m + 1][n + 1] <
        -EPS) return -numeric_limits<DOUBLE>::infinity(); for (int i = 0; i < m; i++) if (B[i]
        == -1) { int s = -1; for (int j = 0; j <= n; j++) if (s == -1 || D[i][j] < D[i][s] ||
        (D[i][j] == D[i][s] && N[j] < N[s])) s = j; Pivot(i, s); } } if (!Simplex(2)) return
        numeric_limits<DOUBLE>::infinity(); x = VD(n); for (int i = 0; i < m; i++) if (B[i] < n)
        x[B[i]] = D[i][n + 1]; /* cerr << "Steepest edge iterations: " << iteration_cnt << "\n"
        ; */ return D[m][n + 1]; }
};
```

9 suffix array

```
#include <bits/stdc++.h>
std::vector<int> sa_is(const std::vector<int>& input) { if (input.size() <= 1) { return
    std::vector<int>(input.size(), 0); } int n = input.size(); int mx =
    *std::max_element(input.begin(), input.end()) + 1; std::cerr << "n: " << n << "   mx: " <<
    mx << "\n" ; /* assert(mx <= n); */ assert(input.back() < *std::min_element(input.begin(),
    input.end() - 1)); std::vector<bool> tp(n + 1); tp[n] = true; for (int i = n - 2; i >= 0;
    i--) { tp[i] = input[i] == input[i + 1] ? tp[i + 1] : (input[i] < input[i + 1]); }
    std::vector<int> suf(n + 1, -1); std::vector<int> bck(mx + 1); for (int val : input)
    bck[val] += 1; std::exclusive_scan(bck.begin(), bck.end(), bck.begin(), 1); std::vector<int>
    ptr(mx + 1); std::copy(bck.begin(), bck.end(), ptr.begin()); suf[0] = n; for (int i = 1; i <
    n; i++) { if (!tp[i - 1] && tp[i]) { suf[--ptr[input[i] + 1]] = i; } } auto induced_sort =
    [&] { std::copy(bck.begin(), bck.end(), ptr.begin()); for (int i = 0; i <= n; i++) { int p =
    suf[i]; if (p > 0 && !tp[p - 1]) { suf[ptr[input[p - 1]]++] = p - 1; } }
    std::copy(bck.begin(), bck.end(), ptr.begin()); for (int i = n; i >= 0; i--) { int p =
    suf[i]; if (p > 0 && tp[p - 1]) { suf[--ptr[input[p - 1] + 1]] = p - 1; } } };
    induced_sort(); int m = 0; std::vector<int> lms_pos(n + 1, -1); for (int i = 1; i <= n; i++)
    { if (!tp[i - 1] && tp[i]) { lms_pos[i] = m++; } } std::vector<int> input2(m); for (int i =
    0, cnt = 0, last = -1; i <= n; i++) { int p = suf[i]; if (p > 0 && !tp[p - 1] && tp[p]) { if
    (last != -1) { if (last == n) { cnt++; } else { for (int j = 0;; j++) { if (p + j == n ||
    last + j == n || input[p + j] != input[last + j]) { cnt++; break; } if (j != 0 && !tp[p + j
    - 1] && tp[p + j]) { break; } } } } last = p; input2[lms_pos[p]] = cnt; } } std::vector<int>
    suf2; if (*std::max_element(input2.begin(), input2.end()) == m - 1) { suf2.assign(m, -1);
    for (int i = 0; i < m; i++) { suf2[input2[i]] = i; } } else { suf2 = sa_is(input2); }
    std::vector<int> ind; ind.swap(lms_pos), ind.clear(); for (int i = 1; i <= n; i++) { if
    (!tp[i - 1] && tp[i]) { ind.push_back(i); } } ind.push_back(n); std::fill(suf.begin(),
    suf.end(), -1); std::copy(bck.begin(), bck.end(), ptr.begin()); suf[0] = n; for (int i = m -
    1; i > 0; i--) { int p = ind[suf2[i]]; suf[--ptr[input[p] + 1]] = p; } induced_sort();
    suf.erase(suf.begin()); return suf; }
std::vector<int> suffix_array(std::vector<int> input) { if (input.size() <= 1) { return
    std::vector<int>(input.size(), 0); } int n = input.size(); int mx =
    *std::max_element(input.begin(), input.end()) + 1; if (mx > n) { std::vector<int> fuck(mx,
    -1); for (int i = 0; i < n; i++) { fuck[input[i]] = 0; } int mx2 = 0; for (int i = 0; i <
    mx; i++) { if (fuck[i] != -1) { fuck[i] = mx2++; } } for (int i = 0; i < n; i++) { input[i]
    = fuck[input[i]]; } mx = mx2; } if (std::all_of(input.begin(), input.end() - 1, [&](int val)
    { return val > input.back(); })) { return sa_is(input); } else { for (auto& i : input) {
    i++; } input.push_back(0); auto suf = sa_is(input); suf.erase(suf.begin()); return suf; } }
```

10 dom tree

```
/* src: https://judge.yosupo.jp/submission/233128 */ #include <bits/stdc++.h>
using namespace std;
struct dominator_tree { int n, t; vector<basic_string<int>> g, rg, bucket; vector<int> arr, par,
    rev, sdom, dom, dsu, label;
    dominator_tree(int n) : g(n), rg(n), bucket(n), arr(n, -1), par(n), rev(n), sdom(n), dom(n),
        dsu(n), label(n), n(n),
    t(0) {}
    void add_edge(int u, int v) { g[u] += v; }
    void dfs(int u) { arr[u] = t; rev[t] = u; label[t] = sdom[t] = dsu[t] = t; t++; for (int w :
        g[u]) { if (arr[w] == -1) { dfs(w); par[arr[w]] = arr[u]; } rg[arr[w]] += arr[u]; } }
    int find(int u, int x = 0) { if (u == dsu[u]) return x ? -1 : u; int v = find(dsu[u], x +
        1); if (v < 0) return u; if (sdom[label[dsu[u]]] < sdom[label[u]]) label[u] =
        label[dsu[u]]; dsu[u] = v; return x ? v : label[u]; }
    vector<int> run(int root) { dfs(root); iota(dom.begin(), dom.end(), 0); for (int i = t - 1;
        i >= 0; i--) { for (int w : rg[i]) sdom[i] = min(sdom[i], sdom[find(w)]); if (i)
        bucket[sdom[i]] += i; for (int w : bucket[i]) { int v = find(w); if (sdom[v] == sdom[w])
        dom[w] = sdom[w]; else dom[w] = v; } if (i > 1) dsu[i] = par[i]; } for (int i = 1; i <
        t; i++) { if (dom[i] != sdom[i]) dom[i] = dom[dom[i]]; } vector<int> outside_dom(n);
        iota(begin(outside_dom), end(outside_dom), 0); for (int i = 0; i < t; i++)
        outside_dom[rev[i]] = rev[dom[i]]; return outside_dom; }
};
int main() { ios::sync_with_stdio(0); cin.tie(0); int n, m, r; cin >> n >> m >> r;
    dominator_tree d(n); while (m--) { int x, y; cin >> x >> y; d.add_edge(x, y); } auto v =
    d.run(r); for (int i = 0; i < n; i++) { if (i != r && v[i] == i) cout << "-1 " ; else cout
    << v[i] << ' ' ; } return 0; }
```

11 poly

```
#include <bits/stdc++.h>
using u32 = uint32_t;
using u64 = uint64_t;
template <u32 mod>
struct MintT { private: u32 m_val;
    static u32 add(u32 a, u32 b) { return a + b - mod * (a + b >= mod); }
    static u32 mul(u32 a, u32 b) { return u64(a) * b % mod; }
    public:
    MintT() : m_val(0) { ; }
    MintT(int64_t x) : m_val((x % mod + mod) % mod) { ; }
    static MintT from_u32_unchecked(u32 val) { MintT m; m.m_val = val; return m; }
    MintT& operator+=(const MintT& other) { return m_val = add(m_val, other.m_val), *this; }
    MintT& operator-=(const MintT& other) { return m_val = add(m_val, mod - other.m_val), *this;
        }
    MintT& operator*=(const MintT& other) { return m_val = mul(m_val, other.m_val), *this; }
    MintT operator-() const { return MintT() - *this; }
    friend MintT operator+(MintT a, const MintT& b) { return MintT(a += b); }
    friend MintT operator-(MintT a, const MintT& b) { return MintT(a -= b); }
    friend MintT operator*(MintT a, const MintT& b) { return MintT(a *= b); }
    MintT power(u64 exp) const { MintT r = 1, b = *this; for (; exp; exp >>= 1) { if (exp & 1) r
        *= b; b *= b; } return r; }
    MintT inverse() const { assert(m_val != 0); return power(mod - 2); }
    static std::vector<MintT> bulk_inverse(const std::vector<MintT>& vec) { std::vector<MintT>
        res(vec.size(), 1); MintT val = 1; for (int i = 0; i < vec.size(); i++) { res[i] *= val,
        val *= vec[i]; } val = val.inverse(); for (int i = 0; i < vec.size(); i++) {
        res.rbegin()[i] *= val, val *= vec.rbegin()[i]; } return res; }
    u32 get_value() const { return m_val; }
    friend bool operator!=(const MintT& a, const MintT& b) { return a.m_val != b.m_val; }
    friend bool operator==(const MintT& a, const MintT& b) { return a.m_val == b.m_val; }
    friend std::istream& operator>>(std::istream& in, MintT& x) { int64_t val; in >> val; x =
        MintT(val); return in; }
    friend std::ostream& operator<<(std::ostream& out, const MintT& x) { return out <<
        x.get_value(); }
};
template <u32 mod>
class NTT { public:
    using mint = MintT<mod>; private: mint pr_root; std::vector<mint> wd, wrd, w_rt, wr_rt;
    static u32 find_pr_root() { std::vector<u32> vec; u32 val = mod - 1; for (u64 i = 2; i * i
        <= val; i++) { if (val % i == 0) { vec.push_back(i); do { val /= i; } while (val % i ==
        0); } } if (val != 1) { vec.push_back(val); } for (u32 i = 2; i < mod; i++) { if
        (std::all_of(vec.begin(), vec.end(),
            [&](u32 q) { return mint(i).power((mod - 1) / q) != 1; })) { return i; }
        }
        assert(false && "pr_root not found" ); }
    public:
    NTT() : pr_root(find_pr_root()) { int lg = __builtin_ctz(mod - 1); wd.assign(lg, 0),
        wrd.assign(lg, 0); w_rt.assign(lg - 1, 0), wr_rt.assign(lg - 1, 0); for (int k = 0; k +
        2 <= lg; k++) { mint a = pr_root.power(mod - 1 >> k + 2); mint b = pr_root.power((mod -
        1 >> k + 2) * ((2 << k) - 2)); w_rt[k] = a, wr_rt[k] = a.inverse(); wd[k] = a *
        b.inverse(), wrd[k] = a.inverse() * b; } }
    private:
    template <bool inverse>
    static void butterfly_x2(mint& a, mint& b, mint w) { mint x = a, y = b; if (!inverse) { y *=
        w, a = x + y, b = x - y; } else { a = x + y, b = (x - y) * w; } }
    public:
    template <bool inverse, bool right_part = false>
    void transform(int lg, mint* data) const { for (int k = inverse ? 0 : lg - 1; inverse ? k <
        lg : k >= 0; inverse ? k++ : k--) { mint wi = right_part ? (inverse ? wr_rt : w_rt)[lg -
        k - 1] : mint(1); for (int i = 0; i < (1 << lg); i += (1 << k + 1)) { for (int j = 0; j
        < (1 << k); j++) { butterfly_x2<inverse>(data[i + j], data[i + (1 << k) + j], wi); } wi
        *= (inverse ? wrd : wd)[__builtin_ctz(~i >> k + 1)]; } } if (inverse) { mint inv =
        mint(mod + 1 >> 1).power(lg); for (int i = 0; i < (1 << lg); i++) { data[i] *= inv; } }
        }
    void expand_ntt(int lg, mint* data) const { std::copy(data, data + (1 << lg), data + (1 <<
        lg)); transform<true>(lg, data + (1 << lg)); transform<false, true>(lg, data + (1 <<
        lg)); }
    void extract_cum(int lg, mint* data, bool odd = false) const { const mint inv2 = mint(mod +
        1 >> 1); if (!odd) { for (int i = 0; i < (1 << lg); i++) { data[i] = (data[2 * i] +
        data[2 * i + 1]) * inv2; } } else { mint wi = 1 * inv2; for (int i = 0; i < (1 << lg);
        i++) { data[i] = (data[2 * i] - data[2 * i + 1]) * wi; wi *= wrd[__builtin_ctz(~i)]; } }
        }
    void convolve_cyclic(int lg, mint* a, mint* b) const { transform<false>(lg, a);
        transform<false>(lg, b); for (int i = 0; i < (1 << lg); i++) { a[i] *= b[i]; }
        transform<true>(lg, a); }
    std::vector<mint> convolve(std::vector<mint> a, std::vector<mint> b) const { if (a.empty()
        || b.empty()) { return {}; } int n = a.size(), m = b.size(), lg = (n == 1 && m == 1) ? 0
        : 32 - __builtin_clz(n + m - 2); if (a.size() * b.size() < int64_t(1 << lg) * lg * 2) {
        std::vector<mint> c(n + m - 1); for (int i = 0; i < n; i++) { for (int j = 0; j < m;
        j++) { c[i + j] += a[i] * b[j]; } } return c; } if (lg > 0 && n + m - 1 == (1 << lg - 1)
        + 1) { mint p = a.back() * b.back(); a.reserve((1 << lg - 1) + 1); a.resize(1 << lg -
        1), b.resize(1 << lg - 1); convolve_cyclic(lg - 1, a.data(), b.data()); a[0] -= p,
        a.push_back(p); return a; } a.resize(1 << lg), b.resize(1 << lg); convolve_cyclic(lg,
        a.data(), b.data()); a.resize(n + m - 1); return a; }
};
namespace polynomial {
    template <u32 mod>
    struct Poly : public std::vector<MintT<mod>> { public:
        using base = std::vector<MintT<mod>>;
        using base::base, base::size, base::resize;
        using mint = MintT<mod>; private: static const NTT<mod> ntt; public:
        mint coeff(size_t ind) const { return ind < this->size() ? this->operator[](ind) :
            mint(); }
        int64_t deg() const { for (int64_t i = size() - 1; i >= 0; i--) { if
            (this->operator[](i) != 0) { return i; } } return -1; }
        friend std::ostream& operator<<(std::ostream& out, const Poly& p) { out << "{" ; for
            (int i = 0; i < p.size(); i++) { if (i != 0) { out << ", " ; } out << p[i]; } out <<
            "}" ; return out; }
        void remove_zeros() { while (size() && this->back() == 0) { this->pop_back(); } }
        friend Poly operator*(const Poly& a, const Poly& b) { int64_t n = a.deg(), m = b.deg();
            if (n == -1 || m == -1) { return {}; } auto p =
            ntt.convolve(std::vector<mint>(a.begin(), a.begin() + n + 1),
            std::vector<mint>(b.begin(), b.begin() + m + 1)); Poly c(p.begin(), p.end());
            c.remove_zeros(); return c; }
        Poly& operator*=(const Poly& other) { return *this = *this * other; }
        Poly operator-() const { Poly a = *this; for (int i = 0; i < a.size(); i++) { a[i] =
            -a[i]; } a.remove_zeros(); return a; }
        Poly& operator+=(const Poly& b) { resize(std::max(size(), b.size())); for (int i = 0; i
            < b.size(); i++) { this->operator[](i) += b[i]; } remove_zeros(); return *this; }
        Poly& operator-=(const Poly& b) { resize(std::max(size(), b.size())); for (int i = 0; i
            < b.size(); i++) { this->operator[](i) -= b[i]; } remove_zeros(); return *this; }
        friend Poly operator+(Poly a, Poly b) { return a += b; }
        friend Poly operator-(Poly a, Poly b) { return a -= b; }
        /* sub x = ax */
        Poly sub_ax(mint a) const { mint p = 1; Poly res = *this; for (int i = 0; i < size();
            i++, p *= a) { res[i] *= p; } return res; }
        Poly div_xk(size_t k) const { return Poly(this->begin() + std::min(size(), k),
            this->end()); }
        Poly mul_xk(size_t k) const { Poly a = *this; a.insert(a.begin(), k, 0); return a; }
        Poly mod_xk(size_t k) const { return Poly(this->begin(), this->begin() +
            std::min(size(), k)); }
```

```
    Poly inv_series(int n) const { Poly a = *this; a.resize(n); Poly b =
        {a.coeff(0).inverse()}; for (int k = 0; (1 << k) < n; k++) { int m = 1 << k; Poly c
        = a.mod_xk(2 * m); b.resize(4 * m); c.resize(4 * m); ntt.template transform<false>(k
        + 2, b.data()); ntt.template transform<false>(k + 2, c.data()); for (int i = 0; i <
        (4 * m); i++) { b[i] *= 2 - b[i] * c[i]; } ntt.template transform<true>(k + 2,
        b.data()); b.resize(2 * m); } b.resize(n); return b; }
    Poly div(Poly b, Poly b_inv = {}) const { Poly a = *this; a.remove_zeros(),
        b.remove_zeros(); assert(b.size()); if (a.size() < b.size()) { return {{}, {}}; }
        std::reverse(a.begin(), a.end()), std::reverse(b.begin(), b.end()); size_t d =
        a.size() - b.size() + 1; if (b_inv.size() < d) { b_inv = b.inv_series(d); } Poly q =
        (a.mod_xk(d) * b_inv.mod_xk(d)).mod_xk(d); q.resize(d); std::reverse(q.begin(),
        q.end()); return q; }
    std::pair<Poly, Poly> divmod(Poly b, const Poly& b_inv = {}) const { Poly q =
        this->div(b, b_inv); Poly r = *this - q * b; assert(r.size() < b.size());
        r.remove_zeros(); return {q, r}; }
    friend Poly operator/(Poly a, Poly b) { return a.div(b); }
    friend Poly operator%(const Poly& a, const Poly& b) { return a.divmod(b).second; }
    Poly power(u64 exp) const { if (exp == 0) { return Poly{1}; } else if (exp & 1) { return
        power(exp - 1) * *this; } else { Poly a = power(exp >> 1); return a * a; } }
    Poly power_mod(u64 exp, const Poly& md, std::shared_ptr<Poly> md_inv = nullptr) const {
        if (exp == 0) { return Poly{1}; } if (md_inv == nullptr || md_inv->size() <
        md.size()) { md_inv = std::make_shared<Poly>(Poly(md.rbegin(),
        md.rend()).inv_series(md.size())); } if (exp & 1) { return (power_mod(exp - 1, md,
        md_inv) * *this).divmod(md, *md_inv).second; } else { Poly a = power_mod(exp >> 1,
        md, md_inv); return (a * a).divmod(md, *md_inv).second; } }
    mint dot(const Poly& b) const { mint res = 0; for (size_t i = 0; i < std::min(size(),
        b.size()); i++) { res += this->operator[](i) * b[i]; } return res; }
    Poly deriv() const { Poly res = *this; for (int i = 0; i < size(); i++) { res[i] *= i; }
        if (res.size()) { res.erase(res.begin()); } res.remove_zeros(); return res; }
    Poly integ() const { Poly res = *this; res.remove_zeros(); mint val = 1; for (int i = 0;
        i < res.size(); i++) { res[i] *= val, val *= (i + 1); } val = val.inverse(); for
        (int i = (int)res.size() - 1; i >= 0; i--) { res[i] *= val, val *= (i + 1); }
        res.insert(res.begin(), 0); res.remove_zeros(); return res; }
    Poly ln(int n) const { if (n <= 1) { return Poly(n); } return (mod_xk(n).deriv() *
        inv_series(n - 1)).mod_xk(n - 1).integ(); }
    Poly exp(int n) const { assert(coeff(0) == 0); Poly b = {1}; for (int k = 0; (1 << k) <
        n; k++) { int m = 1 << k; /* b = (b * (Poly{1} - b.ln(2 * m) + this->mod_xk(2 *
        m))).mod_xk(2 * m); */ Poly b2 = b; Poly c = Poly{1} - b.ln(2 * m) + this->mod_xk(2
        * m); b2.resize(2 * m), c.resize(2 * m), b.resize(2 * m); ntt.convolve_cyclic(k + 1,
        b2.data(), c.data()); for (int i = m; i < 2 * m; i++) { b[i] = b2[i]; } }
        b.resize(n); return b; }
    std::vector<mint> evaluate(const std::vector<mint>& pts) { if (pts.empty()) { return {};
        } int sz = 1; while (sz < pts.size()) sz *= 2; std::vector<Poly> data(2 * sz); for
        (int i = 0; i < pts.size(); i++) { data[sz + i] = Poly({-pts[i], 1}); } for (int i =
        pts.size(); i < sz; i++) { data[sz + i] = Poly({1}); } for (int i = sz - 1; i > 0;
        i--) { data[i] = data[2 * i] * data[2 * i + 1]; } data[1] = *this % data[1]; for
        (int i = 2; i < 2 * sz; i++) { data[i] = data[i >> 1] % data[i]; } std::vector<mint>
        res(pts.size()); for (int i = 0; i < pts.size(); i++) { res[i] = data[sz +
        i].coeff(0); } return res; }
};
template <u32 mod> const NTT<mod> Poly<mod>::ntt;
template <u32 mod>
Poly<mod> interpolate(const std::vector<MintT<mod>>& pts, const std::vector<MintT<mod>>&
    vals) { assert(pts.size() == vals.size()); if (pts.empty()) { return Poly<mod>{}; } int
    sz = 1; while (sz < pts.size()) sz *= 2; std::vector<Poly<mod>> data(2 * sz); for (int i
    = 0; i < pts.size(); i++) { data[sz + i] = Poly<mod>({-pts[i], 1}); } for (int i =
    pts.size(); i < sz; i++) { data[sz + i] = Poly<mod>({1}); } for (int i = sz - 1; i > 0;
    i--) { data[i] = data[2 * i] * data[2 * i + 1]; } std::vector<MintT<mod>> d =
    data[1].deriv().evaluate(pts); d = MintT<mod>::bulk_inverse(d); auto rec = [&](auto rec,
    int i) -> Poly<mod> { if (i >= sz) { if (i - sz < vals.size()) { return Poly<mod>{vals[i
    - sz] * d[i - sz]}; } else { return Poly<mod>{}; } } Poly<mod> a = rec(rec, 2 * i);
        Poly<mod> b = rec(rec, 2 * i + 1); return a * data[2 * i + 1] + b * data[2 * i]; };
        return rec(rec, 1); }
    /* https://arxiv.org/abs/2008.08822 */
    template <u32 mod>
    MintT<mod> bostan_mori(u64 k, Poly<mod> p, Poly<mod> q) { assert(q.coeff(0) != 0);
        using mint = MintT<mod>;
        using poly = Poly<mod>; q.remove_zeros(), p.remove_zeros(); int64_t n = q.deg(); int lg
            = 1; while ((1 << lg) <= 2 * n) { lg++; }
        static const NTT<mod> ntt; q.resize(1 << lg); p.resize(1 << lg); poly r(1 << lg), t(1 <<
            lg); if (n < k) { ntt.template transform<false>(lg, q.data()); ntt.template
            transform<false>(lg, p.data()); while (n < k) { for (int i = 0; i < (1 << lg); i +=
            2) { mint a = p[i], b = p[i + 1], c = q[i], d = q[i + 1]; p[i] = a * d, p[i + 1] = b
            * c, q[i >> 1] = c * d; } ntt.extract_cum(lg - 1, p.data(), k & 1); k >>= 1; if (n <
            k) { ntt.expand_ntt(lg - 1, q.data()); ntt.expand_ntt(lg - 1, p.data()); } }
            ntt.template transform<true>(lg - 1, q.data()); ntt.template transform<true>(lg - 1,
            p.data()); p.resize(k + 1), q.resize(k + 1); p.remove_zeros(), q.remove_zeros(); }
        return (p * q.inv_series(k + 1)).coeff(k); }
    template <u32 mod>
    MintT<mod> kth_linear(u64 k, const Poly<mod>& gen, const Poly<mod>& ch) { /* Poly<mod> r =
        Poly<mod>({0, 1}).power_mod(k, Poly<mod>(ch.rbegin(), ch.rend())); */ /* return
        gen.dot(r); */ int64_t d = ch.deg(); return bostan_mori(k, (gen * ch).mod_xk(d), ch); }
}; /* namespace polynomial */
constexpr u32 mod = 998 '244' 353;
using mint = MintT<mod>;
using poly = polynomial::Poly<mod>;
```

12 multipoint

```
/* https://judge.yosupo.jp/submission/243168 */ #include <bits/stdc++.h> #include <immintrin.h>
    size_t ntt_sum_size = 0;
using u32 = uint32_t;
using u64 = uint64_t;
struct WTF { static constexpr u32 mod = 998 '244' 353; static constexpr u32 pr_root = 3; static
    constexpr int LG = 32; u32 wd[LG], wrd[LG]; u32 w_rt[LG], wr_rt[LG];
    u32 add(u32 a, u32 b) { return a + b - mod * (a + b >= mod); }
    void add_to(u32& a, u32 b) { a = add(a, b); }
    u32 mul(u32 a, u32 b) { return u64(a) * b % mod; }
    u32 power(u32 b, u32 e) { u32 r = 1; for (; e > 0; e >>= 1) { if (e & 1) r = mul(r, b); b =
        mul(b, b); } return r; }
    WTF() { int lg = __builtin_ctz(mod - 1) + 1; for (int i = 0; i < std::min(lg, LG); i++) {
        u32 wi = power(pr_root, mod - 1 >> i + 2); u32 rm = power(pr_root, (mod - 1 >> i + 1) *
        ((1 << i) - 1)); u32 w_dlt = mul(wi, power(rm, mod - 2)); w_rt[i] = wi, wr_rt[i] =
        power(wi, mod - 2); wd[i] = w_dlt, wrd[i] = power(w_dlt, mod - 2); } }
    template <bool transposed>
    void butterfly_x2(u32* ptr_a, u32* ptr_b, u32 w) { u32 a = *ptr_a, b = *ptr_b, a2, b2; if
        (!transposed) { u32 c = mul(b, w); a2 = add(a, c), b2 = add(a, mod - c); } else { a2 =
        add(a, b), b2 = mul(add(a, mod - b), w); } *ptr_a = a2, *ptr_b = b2; }
    template <bool inverse = false, bool right_part = false>
    void transform(int lg, u32* data) { ntt_sum_size += 1 << lg; for (int k = !inverse ? lg - 1
        : 0; !inverse ? k >= 0 : k < lg; !inverse ? k-- : k++) { u32 wi = right_part ? (inverse
        ? wr_rt : w_rt)[lg - 1 - k] : 1; for (int i = 0; i < (1 << lg); i += (1 << k + 1)) { for
        (int j = 0; j < (1 << k); j++) { butterfly_x2<inverse>(data + i + j, data + i + (1 << k)
        + j, wi); } wi = mul(wi, (!inverse ? wd : wrd)[__builtin_ctz(~i >> k + 1)]); } } if
        (inverse) { u32 f = power(mod + 1 >> 1, lg); for (int i = 0; i < (1 << lg); i++) {
        data[i] = mul(data[i], f); } } }
    std::vector<u32> inv_fps(std::vector<u32> vec) { assert(vec.size() && vec[0] != 0); int k =
        0; std::vector<u32> inv = {power(vec[0], mod - 2)}; std::vector<u32> tmp1, tmp2, tmp3;
        while ((1 << k) < vec.size()) { int n = 1 << k; vec.resize(std::max<int>(vec.size(), 2 *
        n)); tmp1.assign(2 * n, 0); std::copy(inv.begin(), inv.begin() + n, tmp1.begin());
        transform<false>(k + 1, tmp1.data()); tmp2.assign(2 * n, 0); std::copy(vec.begin(),
```

```
        vec.begin() + 2 * n, tmp2.begin()); transform<false>(k + 1, tmp2.data()); /* const u32
        fix = power(mod + 1 >> 1, k + 1); */ for (int i = 0; i < 2 * n; i++) { tmp2[i] =
        mul(tmp1[i], tmp2[i]); } transform<true>(k + 1, tmp2.data()); for (int i = 0; i < n;
        i++) { tmp2[i] = 0; } transform<false>(k + 1, tmp2.data()); for (int i = 0; i < 2 * n;
        i++) { tmp1[i] = mul(tmp1[i], add(1, mod - tmp2[i])); } transform<true>(k + 1,
        tmp1.data()); inv.resize(2 * n); std::copy(tmp1.begin() + n, tmp1.begin() + 2 * n,
        inv.begin() + n); k++; } inv.resize(vec.size()); return inv; }
    std::vector<u32> evaluate(std::vector<u32> poly, std::vector<u32> points) { int res_sz =
        points.size(); int n = std::max(poly.size(), points.size()); int lg =
        std::__lg(std::max<int>(n - 1, 1)) + 1; /* * doesn't work for lg = 0 */ poly.resize(1 <<
        lg), points.resize(1 << lg); std::vector<std::vector<u32>> data(lg + 1,
        std::vector<u32>(1 << lg + 1)); for (int i = 0; i < (1 << lg); i++) { data[0][2 * i] =
        add(0, mod + 1 - points[i]); data[0][2 * i + 1] = add(0, mod - 1 - points[i]); } for
        (int k = 0; k < lg; k++) { for (int i = 0; i < (1 << lg + 1); i += 1 << k + 2) { for
        (int j = 0; j < (1 << k + 1); j++) { data[k + 1][i + j] = mul(data[k][i + j], data[k][i
        + (1 << k + 1) + j]); } if (k + 1 != lg) {
                    std::copy(data[k + 1].begin() + i, data[k + 1].begin() + i + (1 << k + 1),
                        data[k + 1].begin() + i + (1 << k + 1)); transform<true>(k + 1, data[k +
                        1].data() + i + (1 << k + 1)); add_to(data[k + 1][i + (1 << k + 1)], mod
                        - 2); transform<false, true>(k + 1, data[k + 1].data() + i + (1 << k +
                        1)); } else { transform<true>(k + 1, data[k + 1].data() + i);
                        add_to(data[k + 1][i], mod - 1); add_to(data[k + 1][i + (1 << k + 1)],
                        1); }
            }
        }
        std::vector<u32> dt = std::move(data[lg]); std::reverse(dt.begin(), dt.begin() + (1 <<
            lg) + 1); dt.resize(1 << lg); dt = inv_fps(dt); std::reverse(dt.begin(), dt.end());
            dt.resize(1 << lg + 1); transform<false>(lg + 1, dt.data()); poly.resize(1 << lg +
            1); std::rotate(poly.begin(), poly.begin() + (1 << lg + 1) - 1, poly.end());
            transform<false>(lg + 1, poly.data()); for (int i = 0; i < (1 << lg + 1); i++) {
            dt[i] = mul(dt[i], poly[i]); }
        for (int k = lg - 1; k >= 0; k--) { for (int i = 0; i < (1 << lg + 1); i += (1 << k +
            2)) { transform<true, true>(k + 1, dt.data() + i + (1 << k + 1)); transform<false>(k
            + 1, dt.data() + i + (1 << k + 1)); for (int j = 0; j < (1 << k + 1); j++) { u32 val
            = add(dt[i + j], mod - dt[i + (1 << k + 1) + j]); dt[i + (1 << k + 1) + j] =
            mul(val, data[k][i + j]); dt[i + j] = mul(val, data[k][i + (1 << k + 1) + j]); } } }
        std::vector<u32> ans(1 << lg); u32 fix = power(mod + 1 >> 1, lg + 1); for (int i = 0; i
            < (1 << lg); i++) { ans[i] = add(dt[2 * i], mod - dt[2 * i + 1]); ans[i] =
            mul(ans[i], fix); }
        ans.resize(res_sz); return ans; }
};
```

13 ******** ntt

```
#include <immintrin.h> #include <algorithm> #include <array> #include <cassert> #include
    <cstdint> #include <cstring> #include <vector>
#pragma GCC target( "avx2,bmi" )
using u32 = uint32_t;
using u64 = uint64_t;
struct Montgomery { u32 mod; /* mod */ u32 mod2; /* 2 * mod */ u32 n_inv; /* n_inv * mod == -1
    (mod 2^32) */ u32 r; /* 2^32 % mod */ u32 r2; /* (2^32)^2 % mod */ Montgomery() = default;
    Montgomery(u32 mod) : mod(mod) { assert(mod % 2 == 1); assert(mod < (1 << 30)); mod2 = 2 *
        mod; n_inv = 1; for (int i = 0; i < 5; i++) { n_inv *= 2 + n_inv * mod; } r = (u64(1) <<
        32) % mod; r2 = u64(r) * r % mod; }
    u32 shrink(u32 val) const { return std::min(val, val - mod); }
    u32 shrink2(u32 val) const { return std::min(val, val - mod2); }
    template <bool strict = true>
    u32 reduce(u64 val) const { u32 res = val + u32(val) * n_inv * u64(mod) >> 32;
        if constexpr (strict) res = shrink(res); return res; }
    template <bool strict = true>
    u32 mul(u32 a, u32 b) const { return reduce<strict>(u64(a) * b); }
    template <bool input_in_space = false, bool output_in_space = false>
    u32 power(u32 b, u32 e) const { if (!input_in_space) b = mul<false>(b, r2); u32 r =
        output_in_space ? this->r : 1; for (; e > 0; e >>= 1) { if (e & 1) r = mul<false>(r, b);
        b = mul<false>(b, b); } return shrink(r); }
};
using i256 = __m256i;
using u32x8 = u32 __attribute__((vector_size(32)));
using u64x4 = u64 __attribute__((vector_size(32)));
u32x8 load_u32x8(const u32* ptr) { return (u32x8)_mm256_load_si256((const i256*)ptr); }
void store_u32x8(u32* ptr, u32x8 vec) { _mm256_store_si256((i256*)ptr, (i256)vec); }
struct Montgomery_simd { u32x8 mod; /* mod */ u32x8 mod2; /* 2 * mod */ u32x8 n_inv; /* n_inv *
    mod == -1 (mod 2^32) */ u32x8 r; /* 2^32 % mod */ u32x8 r2; /* (2^32)^2 % mod */
    Montgomery_simd() = default;
    Montgomery_simd(u32 mod) { Montgomery mt(mod); this->mod = (u32x8)_mm256_set1_epi32(mt.mod);
        this->mod2 = (u32x8)_mm256_set1_epi32(mt.mod2); this->n_inv =
        (u32x8)_mm256_set1_epi32(mt.n_inv); this->r = (u32x8)_mm256_set1_epi32(mt.r); this->r2 =
        (u32x8)_mm256_set1_epi32(mt.r2); }
    u32x8 shrink(u32x8 vec) const { return (u32x8)_mm256_min_epu32((i256)vec,
        _mm256_sub_epi32((i256)vec, (i256)mod)); }
    u32x8 shrink2(u32x8 vec) const { return (u32x8)_mm256_min_epu32((i256)vec,
        _mm256_sub_epi32((i256)vec, (i256)mod2)); }
    u32x8 shrink_n(u32x8 vec) const { return (u32x8)_mm256_min_epu32((i256)vec,
        _mm256_add_epi32((i256)vec, (i256)mod)); }
    u32x8 shrink2_n(u32x8 vec) const { return (u32x8)_mm256_min_epu32((i256)vec,
        _mm256_add_epi32((i256)vec, (i256)mod2)); }
    template <bool strict = true>
    u32x8 reduce(u64x4 x0246, u64x4 x1357) const { u64x4 x0246_ninv =
        (u64x4)_mm256_mul_epu32((i256)x0246, (i256)n_inv); u64x4 x1357_ninv =
        (u64x4)_mm256_mul_epu32((i256)x1357, (i256)n_inv); u64x4 x0246_res =
        (u64x4)_mm256_add_epi64((i256)x0246, _mm256_mul_epu32((i256)x0246_ninv, (i256)mod));
        u64x4 x1357_res = (u64x4)_mm256_add_epi64((i256)x1357,
        _mm256_mul_epu32((i256)x1357_ninv, (i256)mod)); u32x8 res =
        (u32x8)_mm256_or_si256(_mm256_bsrli_epi128((i256)x0246_res, 4), (i256)x1357_res); if
        (strict) res = shrink(res); return res; }
    template <bool strict = true, bool b_use_only_even = false>
    u32x8 mul_u32x8(u32x8 a, u32x8 b) const { u32x8 a_sh = (u32x8)_mm256_bsrli_epi128((i256)a,
        4); u32x8 b_sh = b_use_only_even ? b : (u32x8)_mm256_bsrli_epi128((i256)b, 4); u64x4
        x0246 = (u64x4)_mm256_mul_epu32((i256)a, (i256)b); u64x4 x1357 =
        (u64x4)_mm256_mul_epu32((i256)a_sh, (i256)b_sh); return reduce<strict>(x0246, x1357); }
    template <bool strict = true>
    u64x4 mul_u64x4(u64x4 a, u64x4 b) const { u64x4 pr = (u64x4)_mm256_mul_epu32((i256)a,
        (i256)b); u64x4 pr2 = (u64x4)_mm256_mul_epu32(_mm256_mul_epu32((i256)pr, (i256)n_inv),
        (i256)mod); u64x4 res = (u64x4)_mm256_bsrli_epi128(_mm256_add_epi64((i256)pr,
        (i256)pr2), 4); if (strict) res = (u64x4)shrink((u32x8)res); return res; }
};
class NTT { public: u32 mod, pr_root; private: static constexpr int LG = 32; /* more than enough
    for u32 */ Montgomery mt; Montgomery_simd mts; u32 w[4], wr[4]; u32 wd[LG], wrd[LG]; u64x4
    wt_init, wrt_init; u64x4 wd_x4[LG], wrd_x4[LG]; u64x4 wl_init; u64x4 wld_x4[LG];
    static u32 find_pr_root(u32 mod, const Montgomery& mt) { std::vector<u32> factors; u32 n =
        mod - 1; for (u32 i = 2; u64(i) * i <= n; i++) { if (n % i == 0) { factors.push_back(i);
        do { n /= i; } while (n % i == 0); } } if (n > 1) { factors.push_back(n); } for (u32 i =
        2; i < mod; i++) { if (std::all_of(factors.begin(), factors.end(),
            [&](u32 f) { return mt.power<false, false>(i, (mod - 1) / f) != 1; })) { return i; }
        }
        assert(false && "primitive root not found" ); }
    public: NTT() = default;
    NTT(u32 mod) : mod(mod), mt(mod), mts(mod) { const Montgomery mt = this->mt; const
        Montgomery_simd mts = this->mts; pr_root = find_pr_root(mod, mt); int lg =
        __builtin_ctz(mod - 1); assert(lg <= LG); memset(w, 0, sizeof(w)); memset(wr, 0,
        sizeof(wr)); memset(wd_x4, 0, sizeof(wd_x4)); memset(wrd_x4, 0, sizeof(wrd_x4));
```

```
    memset(wld_x4, 0, sizeof(wld_x4)); std::vector<u32> vec(lg + 1), vecr(lg + 1); vec[lg] =
    mt.power<false, true>(pr_root, mod - 1 >> lg); vecr[lg] = mt.power<true, true>(vec[lg],
    mod - 2); for (int i = lg - 1; i >= 0; i--) { vec[i] = mt.mul<true>(vec[i + 1], vec[i +
    1]); vecr[i] = mt.mul<true>(vecr[i + 1], vecr[i + 1]); } w[0] = wr[0] = mt.r; if (lg >=
    2) { w[1] = vec[2], wr[1] = vecr[2]; if (lg >= 3) { w[2] = vec[3], wr[2] = vecr[3]; w[3]
    = mt.mul<true>(w[1], w[2]); wr[3] = mt.mul<true>(wr[1], wr[2]); } } wt_init =
    (u64x4)_mm256_setr_epi64x(w[0], w[0], w[0], w[1]); wrt_init =
    (u64x4)_mm256_setr_epi64x(wr[0], wr[0], wr[0], wr[1]); wl_init =
    (u64x4)_mm256_setr_epi64x(w[0], w[1], w[2], w[3]); u32 prf = mt.r, prf_r = mt.r; for
    (int i = 0; i < lg - 2; i++) { u32 f = mt.mul<true>(prf, vec[i + 3]), fr =
    mt.mul<true>(prf_r, vecr[i + 3]); prf = mt.mul<true>(prf, vecr[i + 3]), prf_r =
    mt.mul<true>(prf_r, vec[i + 3]); u32 f2 = mt.mul<true>(f, f), f2r = mt.mul<true>(fr,
    fr); wd_x4[i] = (u64x4)_mm256_setr_epi64x(f2, f, f2, f); wrd_x4[i] =
    (u64x4)_mm256_setr_epi64x(f2r, fr, f2r, fr); } prf = mt.r; for (int i = 0; i < lg - 3;
    i++) { u32 f = mt.mul<true>(prf, vec[i + 4]); prf = mt.mul<true>(prf, vecr[i + 4]);
    wld_x4[i] = (u64x4)_mm256_set1_epi64x(f); } }
private: static constexpr int L0 = 3;
int get_low_lg(int lg) const { return lg % 2 == L0 % 2 ? L0 : L0 + 1; }
/* public: */
/* bool lg_available(int lg) { */ /* return L0 <= lg && lg <= __builtin_ctz(mod - 1) +
    get_low_lg(lg); */ /* } */ private:
template <bool transposed, bool trivial = false>
static void butterfly_x2(u32* ptr_a, u32* ptr_b, u32x8 w, const Montgomery_simd& mts) {
    u32x8 a = load_u32x8(ptr_a), b = load_u32x8(ptr_b); u32x8 a2, b2; if (!transposed) { a =
    mts.shrink2(a), b = trivial ? mts.shrink2(b) : mts.mul_u32x8<false, true>(b, w); a2 = a
    + b, b2 = a + mts.mod2 - b; } else { a2 = mts.shrink2(a + b), b2 = trivial ?
    mts.shrink2_n(a - b) : mts.mul_u32x8<false, true>(a + mts.mod2 - b, w); }
    store_u32x8(ptr_a, a2), store_u32x8(ptr_b, b2); }
template <bool transposed, bool trivial = false>
static void butterfly_x4(u32* ptr_a, u32* ptr_b, u32* ptr_c, u32* ptr_d, u32x8 w1, u32x8 w2,
    u32x8 w3, const Montgomery_simd& mts) { u32x8 a = load_u32x8(ptr_a), b =
    load_u32x8(ptr_b), c = load_u32x8(ptr_c), d = load_u32x8(ptr_d); if (!transposed) {
    butterfly_x2<false, trivial>((u32*)&a, (u32*)&c, w1, mts); butterfly_x2<false,
    trivial>((u32*)&b, (u32*)&d, w1, mts); butterfly_x2<false, trivial>((u32*)&a, (u32*)&b,
    w2, mts); butterfly_x2<false, false>((u32*)&c, (u32*)&d, w3, mts); } else {
    butterfly_x2<true, trivial>((u32*)&a, (u32*)&b, w2, mts); butterfly_x2<true,
    false>((u32*)&c, (u32*)&d, w3, mts); butterfly_x2<true, trivial>((u32*)&a, (u32*)&c, w1,
    mts); butterfly_x2<true, trivial>((u32*)&b, (u32*)&d, w1, mts); } store_u32x8(ptr_a, a),
    store_u32x8(ptr_b, b), store_u32x8(ptr_c, c), store_u32x8(ptr_d, d); }
template <bool inverse, bool trivial = false>
void transform_aux(int k, int i, u32* data, u64x4& wi, const Montgomery_simd& mts) const {
    u32x8 w1 = (u32x8)_mm256_shuffle_epi32((i256)wi, 0b00 '00' 00'00); /* only even indices
    will be used */ u32x8 w2 = (u32x8)_mm256_permute4x64_epi64((i256)wi, 0b01 '01' 01'01);
    u32x8 w3 = (u32x8)_mm256_permute4x64_epi64((i256)wi, 0b11 '11' 11'11); for (int j = 0; j
    < (1 << k); j += 8) {
        butterfly_x4<inverse, trivial>(data + i + (1 << k) * 0 + j, data + i + (1 << k) * 1
            + j, data + i + (1 << k) * 2 + j, data + i + (1 << k) * 3 + j, w1, w2, w3, mts);
            }
    wi = mts.mul_u64x4<true>(wi, (inverse ? wrd_x4 : wd_x4)[__builtin_ctz(~i >> k + 2)]); }
public: /* input in [0, 4 * mod) */ /* output in [0, 4 * mod) */ /* data must be 32-byte
    aligned */
void transform_forward(int lg, u32* data) const { const Montgomery_simd mts = this->mts;
    const int L = get_low_lg(lg); /* for (int k = lg - 2; k >= L; k -= 2) { */ /* u64x4 wi =
    wt_init; */ /* transform_aux<false, true>(k, 0, data, wi, mts); */ /* for (int i = (1 <<
    k + 2); i < (1 << lg); i += (1 << k + 2)) { */ /* transform_aux<false>(k, i, data, wi,
    mts); */ /* } */ /* } */ if (L < lg) { const int lc = (lg - L) / 2; u64x4 wi_data[LG /
    2]; std::fill(wi_data, wi_data + lc, wt_init); for (int k = lg - 2; k >= L; k -= 2) {
    transform_aux<false, true>(k, 0, data, wi_data[k - L >> 1], mts); } for (int i = 1; i <
    (1 << lc * 2 - 2); i++) { int s = __builtin_ctz(i) >> 1; for (int k = s; k >= 0; k--) {
    transform_aux<false>(2 * k + L, i * (1 << L + 2), data, wi_data[k], mts); } } } }
/* input in [0, 2 * mod) */ /* output in [0, mod) */ /* data must be 32-byte aligned */
template <bool mul_by_sc = false>
void transform_inverse(int lg, u32* data, /* as normal number */ u32 sc = u32()) const {
    const Montgomery_simd mts = this->mts; const int L = get_low_lg(lg); /* for (int k = L;
    k + 2 <= lg; k += 2) { */ /* u64x4 wi = wrt_init; */ /* transform_aux<true, true>(k, 0,
    data, wi, mts); */ /* for (int i = (1 << k + 2); i < (1 << lg); i += (1 << k + 2)) { */
    /* transform_aux<true>(k, i, data, wi, mts); */ /* } */ /* } */ if (L < lg) { const int
    lc = (lg - L) / 2; u64x4 wi_data[LG / 2]; std::fill(wi_data, wi_data + lc, wrt_init);
    for (int i = 0; i < (1 << lc * 2 - 2); i++) { int s = __builtin_ctz(~i) >> 1; if (i + 1
    == (1 << 2 * s)) { s--; } for (int k = 0; k <= s; k++) {
                transform_aux<true>(2 * k + L, (i + 1 - (1 << 2 * k)) * (1 << L + 2), data,
                    wi_data[k], mts); }
            if (i + 1 == (1 << 2 * (s + 1))) { s++;
                transform_aux<true, true>(2 * s + L, (i + 1 - (1 << 2 * s)) * (1 << L + 2),
                    data, wi_data[s], mts); }
        }
    }
    const Montgomery mt = this->mt; u32 f = mt.power<false, true>(mod + 1 >> 1, lg - L);
    if constexpr (mul_by_sc) f = mt.mul<true>(f, mt.mul<false>(mt.r2, sc)); u32x8 f_x8 =
        (u32x8)_mm256_set1_epi32(f); for (int i = 0; i < (1 << lg); i += 8) {
        store_u32x8(data + i, mts.mul_u32x8<true, true>(load_u32x8(data + i), f_x8)); }
}
private: /* input in [0, 4 * mod) */ /* output in [0, 2 * mod) */ /* multiplies mod (x^2^L -
    w) */
template <int L, int K, bool remove_montgomery_reduction_factor = true> /* !!! O3 is crucial
    here !!! */ __attribute__((optimize( "O3" ))) static void
aux_mul_mod_x2L(const u32* a, const u32* b, u32* c, const std::array<u32x8, K>& ar_w, const
    Montgomery_simd& mts) { static_assert(L >= 3); /* static_assert(L == L0 || L == L0 + 1);
    */
    constexpr int n = 1 << L; alignas(64) u32 aux_a[K][n]; alignas(64) u64 aux_b[K][n * 2];
        for (int k = 0; k < K; k++) { for (int i = 0; i < n; i += 8) { u32x8 ai =
        load_u32x8(a + n * k + i);
            if constexpr (remove_montgomery_reduction_factor) { ai = mts.mul_u32x8<true,
                true>(ai, mts.r2); } else { ai = mts.shrink(mts.shrink2(ai)); }
            store_u32x8(aux_a[k] + i, ai); u32x8 bi = load_u32x8(b + n * k + i); u32x8 bi_0
                = mts.shrink(mts.shrink2(bi)); u32x8 bi_w = mts.mul_u32x8<true, true>(bi,
                ar_w[k]);
            store_u32x8((u32*)(aux_b[k] + i + 0),
                (u32x8)_mm256_permutevar8x32_epi32((i256)bi_w, _mm256_setr_epi64x(0, 1, 2,
                3)));
            store_u32x8((u32*)(aux_b[k] + i + 4),
                (u32x8)_mm256_permutevar8x32_epi32((i256)bi_w, _mm256_setr_epi64x(4, 5, 6,
                7)));
            store_u32x8((u32*)(aux_b[k] + n + i + 0),
                (u32x8)_mm256_permutevar8x32_epi32((i256)bi_0, _mm256_setr_epi64x(0, 1, 2,
                3)));
            store_u32x8((u32*)(aux_b[k] + n + i + 4),
                (u32x8)_mm256_permutevar8x32_epi32((i256)bi_0, _mm256_setr_epi64x(4, 5, 6,
                7))); }
    }
    u64x4 aux_ans[K][n / 4]; memset(aux_ans, 0, sizeof(aux_ans)); for (int i = 0; i < n;
        i++) { for (int k = 0; k < K; k++) { u64x4 ai =
        (u64x4)_mm256_set1_epi32(aux_a[k][i]); for (int j = 0; j < n; j += 4) { u64x4 bi =
        (u64x4)_mm256_loadu_si256((i256*)(aux_b[k] + n - i + j)); aux_ans[k][j / 4] += /*
        64-bit addition */ (u64x4)_mm256_mul_epu32((i256)ai, (i256)bi); } } if (i >= 8 && (i
        & 7) == 7) { for (int k = 0; k < K; k++) { for (int j = 0; j < n; j += 4) {
        aux_ans[k][j / 4] = (u64x4)mts.shrink2((u32x8)aux_ans[k][j / 4]); } } } }
    for (int k = 0; k < K; k++) { for (int i = 0; i < n; i += 8) { u64x4 c0 = aux_ans[k][i /
        4], c1 = aux_ans[k][i / 4 + 1]; u32x8 res = (u32x8)_mm256_permutevar8x32_epi32(
        (i256)mts.reduce<false>(c0, c1), _mm256_setr_epi32(0, 2, 4, 6, 1, 3, 5, 7));
        store_u32x8(c + k * n + i, mts.shrink2(res)); } }
}
```

```
    template <int L, bool remove_montgomery_reduction_factor = true>
    void aux_mul_mod_full(int lg, const u32* a, const u32* b, u32* c) const {
        constexpr int sz = 1 << L; const Montgomery_simd mts = this->mts; int cnt = 1 << lg - L;
            if (cnt == 1) { aux_mul_mod_x2L<L, 1, remove_montgomery_reduction_factor>(a, b, c,
            {mts.r}, mts); return; }
        if (cnt <= 8) { for (int i = 0; i < cnt; i += 2) { u32x8 wi =
            (u32x8)_mm256_set1_epi32(w[i / 2]);
                aux_mul_mod_x2L<L, 2, remove_montgomery_reduction_factor>( a + i * sz, b + i *
                    sz, c + i * sz, {wi, (mts.mod - wi)}, mts); }
            return; }
        u64x4 wi = wl_init; for (int i = 0; i < cnt; i += 8) { u32x8 w_ar[4] = {
            (u32x8)_mm256_permute4x64_epi64((i256)wi, 0b00 '00' 00'00),
            (u32x8)_mm256_permute4x64_epi64((i256)wi, 0b01 '01' 01'01),
            (u32x8)_mm256_permute4x64_epi64((i256)wi, 0b10 '10' 10'10),
            (u32x8)_mm256_permute4x64_epi64((i256)wi, 0b11 '11' 11'11), };
            if constexpr (L == L0) { for (int j = 0; j < 8; j += 4) {
                    aux_mul_mod_x2L<L, 4, remove_montgomery_reduction_factor>( a + (i + j) * sz,
                        b + (i + j) * sz, c + (i + j) * sz, {w_ar[j / 2], mts.mod - w_ar[j / 2],
                        w_ar[j / 2 + 1], mts.mod - w_ar[j / 2 + 1]}, mts); }
            } else { for (int j = 0; j < 8; j += 2) {
                    aux_mul_mod_x2L<L, 2, remove_montgomery_reduction_factor>( a + (i + j) * sz,
                        b + (i + j) * sz, c + (i + j) * sz, {w_ar[j / 2], mts.mod - w_ar[j /
                        2]}, mts); }
            }
            wi = mts.mul_u64x4<true>(wi, wld_x4[__builtin_ctz(~i >> 3)]); }
    }
    public:
    template <bool remove_montgomery_reduction_factor = true>
    void aux_dot_mod(int lg, const u32* a, const u32* b, u32* c) const { int L = get_low_lg(lg);
        if (L == L0) { aux_mul_mod_full<L0, remove_montgomery_reduction_factor>(lg, a, b, c); }
        else { aux_mul_mod_full<L0 + 1, remove_montgomery_reduction_factor>(lg, a, b, c); } }
    /* lg must be greater than or equal to 3 */ /* a, b must be 32-byte aligned */
    void convolve_cyclic(int lg, u32* a, u32* b) const { transform_forward(lg, a);
        transform_forward(lg, b); aux_dot_mod(lg, a, b, a); transform_inverse(lg, a); }
};
```

14 cached compile

```
import os, subprocess
def cached_compile(path):
    if not os.path.exists(path) or os.path.getmtime(path) < os.path.getmtime(f"{path}.cpp"):
        print(f"compiling {path}...", end=" ", flush=True)
        ret = subprocess.run(f"g++ {path}.cpp -o {path} -std=c++20 -O2", shell=True).returncode
        if ret != 0:
            print("fuck")
            exit(1)
        print("compiled", flush=True)
```

15 suff automaton

```
#include <bits/stdc++.h>
struct Node { int32_t go[26]; int32_t suf_link = -1; int32_t parent = -1; int32_t count = 1;
    int32_t min_len = 1 '000' 000 '000, max_len = -1' 000 '000' 000;
    Node() { for (int32_t i = 0; i < 26; i++) go[i] = -1; }
    void copy_go(Node& node) { for (int32_t i = 0; i < 26; i++) go[i] = node.go[i]; }
};
int32_t next = 1;
int32_t extend(int32_t last, int32_t ch, Node* nodes) { int32_t b = next++; nodes[b].parent =
    last; for (int32_t a = last; a > -1; a = nodes[a].suf_link) { if (nodes[a].go[ch] == -1) {
    nodes[a].go[ch] = b; continue; } int32_t c = nodes[a].go[ch]; if (nodes[c].parent == a) {
    nodes[b].suf_link = c; return b; } int32_t clone = next++; nodes[clone].copy_go(nodes[c]);
    nodes[clone].parent = a; nodes[clone].suf_link = nodes[c].suf_link; nodes[c].suf_link =
    clone; nodes[b].suf_link = clone; nodes[clone].count = 0; for (; a > -1; a =
    nodes[a].suf_link) if (nodes[a].go[ch] == c) nodes[a].go[ch] = clone; else break; return b;
    } nodes[b].suf_link = 0; return b; }
```

16 incremental mst

```
/* https://codeforces.com/blog/entry/158509 */ const int inf = 1e9;
struct BottleneckTree { std::vector<int> link, cost; /* all inserted costs must be < inf */
    BottleneckTree(int n) : link(n, 0), cost(n, inf) { link[0] = -1; cost[0] = inf + 1; }
    void rot(int x) { int p = link[x], g = link[p]; swap(cost[x], cost[p]); link[x] = g, link[p]
        = x; }
    void skip(int x) { link[x] = link[link[x]]; }
    void access(int x) { for (int p = link[x]; p != -1; p = link[x]) { int g = link[p]; if
        (cost[x] <= cost[p]) rot(x); /* (A) */ else if (cost[p] > cost[g]) skip(p), skip(x); /*
        (B) */ else if (cost[g] <= cost[x]) rot(p), skip(x); /* (C) */ else skip(x); /* (D) */ }
        }
    int Query(int x, int y) { /* x != y; inf if disconnected */ access(x); access(y); return
        cost[x]; }
    int AddEdge(int x, int y, int c) { /* x != y */ access(x); access(y); if (cost[x] > c)
        swap(cost[x], c); return c; }
};
```

17 Hilbert order fast

```
#pragma GCC target( "bmi2" ) #include <immintrin.h> #include <cstdint>
uint64_t interleave(uint64_t x) { /* x = (x | (x << 16))& 0x0000 'FFFF' 0000
    'FFFF; x = (x | (x << 8)) & 0x00FF' 00FF '00FF' 00FF; */ /* x = (x | (x << 4)) & 0x0F0F
    '0F0F' 0F0F '0F0F; x = (x | (x << 2)) & 0x3333' 3333 '3333' 3333; */ /* x = (x | (x << 1)) &
    0x5555 '5555' 5555'5555; */ const uint64_t mask = 0x5555 '5555' 5555'5555; x = _pdep_u64(x,
    mask) | _pdep_u64(x >> 32, mask) << 1; return x; } /* http://threadlocalmutex.com/?p=126 //
    for 64-bit indicies */
uint64_t hilbertXYToIndex_logarithmic(uint32_t x, uint32_t y) { uint32_t A, B, C, D; /* Initial
    prefix scan round, prime with x and y */ { uint32_t a = x ^ y, b = 0xFFFF
    'FFFF ^ a, c = 0xFFFF' FFFF ^ (x | y), d = x & (y ^ 0xFFFF'FFFF); A = a | (b >> 1), B = (a
    >> 1) ^ a, C = ((c >> 1) ^ (b & (d >> 1))) ^ c; D = ((a & (c >> 1)) ^ (d >> 1)) ^ d; } {
    uint32_t a = A, b = B, c = C, d = D; A = ((a & (a >> 2)) ^ (b & (b >> 2))), B = ((a & (b >>
    2)) ^ (b & ((a ^ b) >> 2))); C ^= ((a & (c >> 2)) ^ (b & (d >> 2))), D ^= ((b & (c >> 2)) ^
    ((a ^ b) & (d >> 2))); } { uint32_t a = A, b = B, c = C, d = D; A = ((a & (a >> 4)) ^ (b &
    (b >> 4))), B = ((a & (b >> 4)) ^ (b & ((a ^ b) >> 4))); C ^= ((a & (c >> 4)) ^ (b & (d >>
    4))), D ^= ((b & (c >> 4)) ^ ((a ^ b) & (d >> 4))); } { uint32_t a = A, b = B, c = C, d = D;
    A = ((a & (a >> 8)) ^ (b & (b >> 8))), B = ((a & (b >> 8)) ^ (b & ((a ^ b) >> 8))); C ^= ((a
    & (c >> 8)) ^ (b & (d >> 8))), D ^= ((b & (c >> 8)) ^ ((a ^ b) & (d >> 8))); } /* Final
    round and projection */{ uint32_t a = A, b = B, c = C, d = D; C ^= ((a & (c >> 16)) ^ (b &
    (d >> 16))), D ^= ((b & (c >> 16)) ^ ((a ^ b) & (d >> 16))); } uint32_t a = C ^ (C >> 1), b
    = D ^ (D >> 1); /* Undo transformation prefix scan */ uint32_t i0 = x ^ y, i1 = b |
    (0xFFFF'FFFF ^ (i0 | a)); /* Recover index bits */ return (interleave(i1) << 1) |
    interleave(i0); }
```

18 Sierpinski order

```
int64_t get_cum_rec(int x, int y, int lg = 30) { if (lg == 0) return 0; x &= (1 << lg) - 1, y &=
    (1 << lg) - 1; const int i = lg - 1, int sz = 1 << i; const int64_t all = int64_t(1) << 2 *
    i; int a = x >> i & 1, b = y >> i & 1, x2 = x & sz - 1, y2 = y & sz - 1; int64_t res = 0; if
    (a == 0 && b == 0) { if (x > y || x == y && x != 0 && x % 2 == 0) res += all * 3; return res
```

```
    + get_cum_rec(x2, y2, lg - 1); } res += all / 2 + 1; if (a == 1 && b == 1) { res += all; if
    (x2 == 0 && y2 == 0) return res + (all - 1); return res + get_cum_rec(x2, y2, lg - 1) - 1; }
    else if (a == 0 && b == 1) { if (x2 == sz - 1 && y2 == 0) return res; return res + all -
    get_cum_rec(sz - 1 - x2, y2, lg - 1); } else { res += all * 2; if (x2 == 0 && y2 == sz - 1)
    return res; return res + all - get_cum_rec(x2, sz - 1 - y2, lg - 1); } }
```

19 Lindon (min cyclic shift)

```
/* Lyndon: // String is simple, if it is smaller than all its suffixes */ /* s = s_1 s_2 ...
    s_k, s_1 >= s_2 >= ... >= s_k, s_i is simple */
std::string min_cyclic_shift (std::string s) { s += s; int n = (int) s.length(), i=0, ans=0;
    while (i < n/2) { ans = i; int j=i+1, k=i; while (j < n && s[k] <= s[j]) { if (s[k] < s[j])
    {k = i;} else {++k;} ++j; } while (i <= k) i += j - k; } return s.substr(ans, n/2); }
```

20 Gomory-Hu

```
/* input: undirected graph */ /* output: tree in form [p_i, w_i], res[0] is empty */ /* runs
    dinic n-1 times */ /* mincut in this tree <=> mincut in original graph */ /* g - (u, v, w)
    */
vector<pair<int, int>> gomory_hu(int n, vector<tuple<int, int, int>> const &g) { Dinic f(n, 0,
    0); for (auto [v, u, w] : g) f.add(v, u, w, w); vector<pair<int, int>> res(n); vector<int>
    pr(n, 0); for (int i = 1; i < n; i++) { for (auto &j : f.ed) j.f = 0; f.st = i; f.fin =
    pr[i]; auto flow = f.get(); vector<bool> cut(n); for (int j = 0; j < n; j++) cut[j] =
    (f.ds[j] != -1); for (int j = i + 1; j < n; j++) { if (cut[j] == cut[i] && pr[j] == pr[i])
    pr[j] = i; } res[i] = {pr[i], flow}; } return res; }
```